4 System parameters

4.1 Introduction

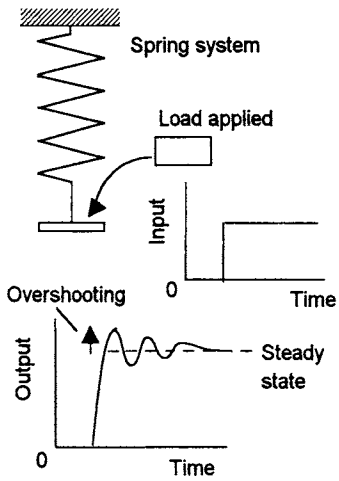

Figure 4.1 *A spring system with an output to a step input which takes time to reach the steady state value and shows overshooting*

When a system is subject to, say, a unit step input it may give an output which eventually settles down to some steady state response. The response that it gives before settling down to this steady state is called its *transient response*. This chapter is about the parameters used to specify the transient response of systems and whether the transients lead to unstable systems.

For example, if we have a spring system (Figure 4.1) and suddenly apply a load to it, it has a transient response which results in it taking some time to reach its steady state value and also it is likely to overshoot the steady state value before it finally settles down to the steady state value. What factors can we change with the spring system in order to get it to respond more quickly to an input and also to minimise the overshooting? These are questions that are often posed for control systems. As another illustration, consider a control system used with an automatic machine to position a workpiece before some machining operation, we need to know how fast the system will respond to an input signal and position the item in the required position and will the system be like the spring system when a load is applied to it and overshooting of the required position occur. Overshooting is undesirable in such a situation and so, if it occurred in such a control system, we would need to consider what steps can be taken to eliminate it. Parameters are used as a way of specifying how fast a system will respond to an input and how quickly it will settle down to its steady state value.

With the above spring system, the result of applying a load is that, after some oscillations with ever decreasing amplitude, the transients die away and the system settles down to a stead state value. The system, is said to be stable. If, however, the oscillations had continued with ever increasing amplitude, then no steady state value would have been reached and the system would be unstable. This chapter takes a brief look at the conditions for stability of systems.

4.2 First order systems

A first order system has a transfer function of the form:

$$G(s) = \frac{k}{\tau s + 1}$$

where k is the steady state gain and τ the time constant. When a unit step input is applied to such a system, the output y is:

$$y = k(1 - e^{-t/\tau})$$

When the time $t = \tau$ then we have $y = k(1 - e^{-1}) = 0.63k$. Thus, the *time constant* τ for a first order system when subject to a step input is the time taken for the output to reach 0.63, of the steady state value.

For a first order system, the parameters used to specify the transient performance are:

1 *Delay time*
The delay time t_d is the time required for the output response to reach 50% of its steady state value. Thus, since k is the final value, the time taken to reach 50% of this value is given by:

$$\tfrac{1}{2}k = k(1 - e^{-t_d/\tau})$$

$$\tfrac{1}{2} = e^{-t_d/\tau}$$

$$\ln 2 = \frac{t_d}{\tau}$$

$$t_d = \tau \ln 2$$

2 *Rise time*
The rise time t_r is the time required for the output to rise from 10% to 90% of its steady state value. Note that the specification is not always in terms of 10% to 90%, sometimes it is 0% to 100%. Since k is the steady state value then the time taken to reach 10% of that value is:

$$\frac{10}{100}k = k(1 - e^{-t_{10}/\tau})$$

$$\frac{1}{10} = e^{-t_{10}/\tau}$$

$$\ln 10 = \frac{t_{10}}{\tau}$$

$$t_{10} = \tau \ln 10$$

The time taken to reach 90% of the steady state value is given by:

$$\frac{90}{100}k = k(1 - e^{-t_{90}/\tau})$$

$$\frac{9}{10} = e^{-t_{90}/\tau}$$

$$\ln 10 - \ln 9 = \frac{t_{90}}{\tau}$$

$$t_{90} = \tau \ln 10 - \tau \ln 9$$

Hence the rise time is:

$$t_r = t_{90} - t_{10} = \tau \ln 9$$

Example

Determine the delay time and the rise time for a first order system with the transfer function:

$$G(s) = \frac{3}{2s+1}$$

Comparing the transfer function with $k/(\tau s + 1)$ indicates that the steady state gain is 3 and the time constant is 2 s. Thus, the delay time is:

$$t_{10} = \tau \ln 10 = 2 \ln 10 = 4.6 \text{ s}$$

The rise time is:

$$t_r = t_{90} - t_{10} = \tau \ln 9 = 2 \ln 9 = 4.4 \text{ s}$$

Example

A mercury-in-glass thermometer acts as a first order system with an input of temperature and an output of the mercury position against a scale. The thermometer is initially at 0°C and is then suddenly placed in water at 100°C. After 80 s the thermometer reads 98°C. Determine (a) the time constant, (b) the delay time, (c) the rise time.

(a) For such a system the output θ is related to the input by the equation:

$$\theta = 100(1 - e^{-t/\tau})$$

Hence:

$$98 = 100(1 - e^{-80/\tau})$$

$$0.2 = e^{-80/\tau}$$

$$\ln 0.2 = -\frac{80}{\tau}$$

Hence, the time constant τ is 49.7 s.
(b) The delay time is:

$$t_{10} = \tau \ln 10 = 49.7 \ln 10 = 114.4 \text{ s}$$

(c) The rise time is:

$$t_r = t_{90} - t_{10} = \tau \ln 9 = 49.7 \ln 9 = 109.2 \text{ s}$$

4.3 Second order systems

A second order system has a transfer function of the form:

$$G(s) = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The form the output takes for a unit step input depends on whether the damping factor ζ is less than, equal to or greater than 1, the corresponding outcomes being underdamped, critically damped and overdamped (see Section 3.6).

For the underdamped oscillations of a system we have the output y given by:

$$y = k\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_n \sqrt{(1-\zeta^2)}\, t + \phi\right)\right]$$

with the damped natural frequency ω given by:

$$\omega = \omega_n \sqrt{(1-\zeta^2)}$$

We can write the above equation for the output in what is often a more convenient form. Since $\sin(A + B) = \sin A \cos B + \cos A \sin B$, the sine term can be written as:

$$\sin(\omega t + \phi) = \sin \omega t \cos \phi + \cos \omega t \sin \phi$$

and since ϕ is a constant:

$$\sin(\omega t + \phi) = P \sin \omega t + Q \cos \omega t$$

where P and Q are constants. Thus the output can be written as:

$$y = k\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}(P \sin \omega t + Q \cos \omega t)\right]$$

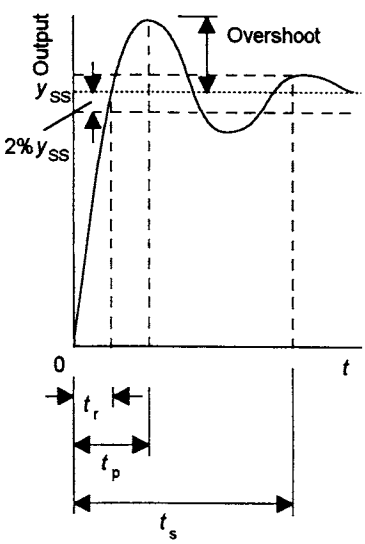

Figure 4.2 *Step response of an under-damped system*

The performance of an underdamped second order system to a unit step input (Figure 4.2) can be specified by:

1 *Rise time*

The rise time t_r is the time taken for the response x to rise from 0 to the steady-state value y_{SS}. This is the time for the oscillating response to complete a quarter of a cycle, i.e. $\frac{1}{2}\pi$. Thus:

$$\omega t_r = \tfrac{1}{2}\pi$$

We can thus reduce the rise time by increasing the damped natural frequency, this value being determined by the undamped natural angular frequency and the damping factor, i.e.

$$t_r = \frac{\pi}{2\omega_n \sqrt{1-\zeta^2}}$$

The rise time is sometimes specified as the time taken for the response to rise from 10% to 90% of the steady state value.

2 *Peak time*

The peak time t_p is the time taken for the response to rise from 0 to the first peak value. This is the time for the oscillating response to complete one half-cycle, i.e. π . Thus:

$$\omega t_p = \pi$$

and so we can write:

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

When ζ is 1 then the peak time becomes infinite; this indicates that at critical damping the steady state value is never reached but only approached asymptotically.

3 *Overshoot*

The overshoot is the maximum amount by which the response overshoots the steady state value and is thus the amplitude of the first peak. The overshoot is often written as a percentage of the steady state value.

The steady state value is when t tends to infinity and thus $y_{SS} = k$. Since $y = 0$ when $t = 0$ then, since $e^0 = 1$, we have:

$$0 = k\left[1 - \frac{1}{\sqrt{1-\zeta^2}}(0+Q)\right]$$

and so $Q = \sqrt{(1-\zeta^2)}$.

The overshoot occurs at $\omega t = \pi$ and thus:

$$y = k\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}(P\sin\omega t + Q\cos\omega t)\right]$$

becomes:

$$y = y_{SS}\left[1 - \frac{e^{-\zeta\omega_n \pi/\omega}}{\sqrt{1-\zeta^2}}(0-Q)\right]$$

The overshoot is the difference between the output at that time and the steady-state value. Hence:

$$\text{overshoot} = y_{SS}\frac{e^{-\zeta\omega_n\pi/\omega}}{\sqrt{1-\zeta^2}}Q = y_{SS}\, e^{-\zeta\omega_n\pi/\omega}$$

Since $\omega = \omega_n\sqrt{(1-\zeta^2)}$ then we can write:

$$\text{Overshoot} = y_{ss} \exp\left(\frac{-\zeta\omega_n\pi}{\omega_n\sqrt{1-\zeta^2}}\right)$$

$$= y_{ss} \exp\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)$$

Expressed as a percentage of y_{ss}:

$$\text{percentage overshoot} = \exp\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right) \times 100\%$$

Note that the overshoot does not depend on the natural frequency of the system but only on the damping factor. As the damping factor approaches 1 so the percentage overshoot approaches zero. Table 4.1 gives values of the percentage overshoot for particular damping ratios.

Table 4.1 *Percentage peak overshoot*

Damping ratio	Percentage overshoot
0.2	52.7
0.3	37.2
0.4	25.4
0.5	16.3
0.6	9.5
0.7	4.6
0.8	1.5
0.9	0.2

4 *Subsidence ratio*

An indication of how fast oscillations decay is provided by the *subsidence ratio* or *decrement*. This is the amplitude of the second overshoot divided by the amplitude of the first overshoot. The first overshoot occurs when we have $\omega t = \pi$ and so:

$$\text{first overshoot} = y_{ss} \exp\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)$$

The second overshoot occurs when $\omega t = 3\pi$ and so:

$$\text{second overshoot} = y_{ss} \exp\left(\frac{-3\zeta\pi}{\sqrt{1-\zeta^2}}\right)$$

Thus the subsidence ratio is given by:

$$\text{subsidence ratio} = \frac{\text{second overshoot}}{\text{first overshoot}} = \exp\left(\frac{-2\zeta\pi}{\sqrt{1-\zeta^2}}\right)$$

5 *Settling time*

The settling time t_s is used as a measure of the time taken for the oscillations to die away. It is the time taken for the response to fall within and remain within some specified percentage of the steady-state value (see Figure 4.1). Thus for the 2% settling time, the amplitude of the oscillation should fall to be less than 2% of y_{SS}. We have:

$$y = k\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}(P\sin\omega t + Q\cos\omega t)\right]$$

with $y_{SS} = k$, $\omega = \omega_n\sqrt{(1-\zeta^2)}$ and, as derived earlier in item 3, $Q = \sqrt{(1-\zeta^2)}$. The amplitude of the oscillation is $(y - y_{SS})$ when y is a maximum value. The maximum values occur when ωt is some multiple of π and thus we have $\cos\omega t = 1$ and $\sin\omega t = 0$. For the 2% settling time, the settling time t_s is when the maximum amplitude is 2% of y_{SS}, i.e. $0.02y_{SS}$. Thus:

$$0.02y_{SS} = y_{SS}\,e^{-\zeta\omega_n t_s}$$

Taking logarithms gives $\ln 0.02 = -\zeta\omega_n t_s$ and since $\ln 0.02 = -3.9$ or approximately 4 then:

$$t_s = \frac{4}{\zeta\omega_n}$$

The above is the value of the settling time if the specified percentage is 2%. If the specified percentage is 5% the equation becomes

$$t_s = \frac{3}{\zeta\omega_n}$$

6 *Number of oscillations to settling time*

The time taken to complete one cycle, i.e. the periodic time, is $1/f$, where f is the frequency, and since $\omega = 2\pi f$ then the time to complete one cycle is $2\pi/f$. In a settling time of t_s the number of oscillations that occur is:

$$\text{number of oscillations} = \frac{\text{settling time}}{\text{periodic time}}$$

and thus for a settling time defined for 2% of the steady-state value:

$$\text{number of oscillations} = \frac{4/\zeta\omega_n}{2\pi/\omega}$$

Since $\omega = \omega_n\sqrt{(1-\zeta^2)}$, then:

$$\text{number of oscillations} = \frac{2\omega_n\sqrt{1-\zeta^2}}{\pi\zeta\omega_n} = \frac{2}{\pi}\sqrt{\frac{1}{\zeta^2}-1}$$

In designing a system the following are the typical points that are considered:

1 For a rapid response, i.e. small rise time, the natural frequency must be large. Figure 4.3 shows the types of response obtained to a unit step input to systems having the same damping factor of 0.2 but different natural angular frequencies. The response time with the natural angular frequency of 10 rad/s, damped frequency 9.7 rad/s, is much higher than that with a natural angular frequency of 1 rad/s, damped frequency 0.97 rad/s.

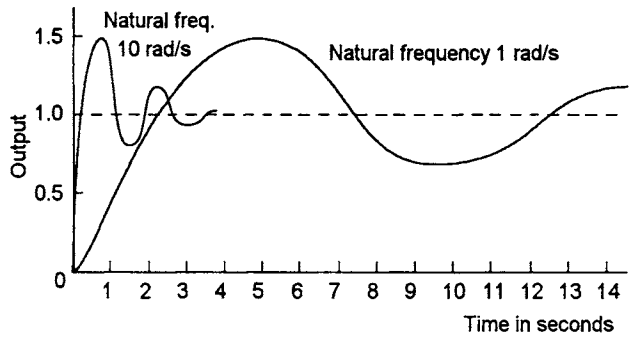

Figure 4.3 *Response of a unit gain second order system to a unit step input, the damping factor being the same for both responses*

2 The damping factor is typically in the range 0.4 to 0.8 since smaller values give an excessive overshoot and a large number of oscillations before the system settles down. Larger values render the system sluggish since they increase the response time. Though, in some systems where no overshoot can be tolerated, a high value of damping factor may have to be used. Figure 4.4 shows the effect on the response of a second order system of a change of damping factor when the natural angular frequency remains unchanged.

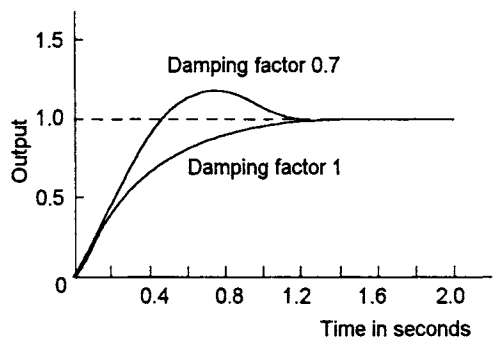

Figure 4.4 *Response of a unit gain second order system to a unit step input, the natural angular frequency being the same for both responses*

Example

A second order system has a natural angular frequency of 2.0 rad/s and a damped frequency of 1.8 rad/s. What are its (a) damping factor, (b) 100% rise time, (c) percentage overshoot, (c) 2% settling time, and (d) the number of oscillations within the 2% settling time?

(a) Since $\omega = \omega_n\sqrt{(1-\zeta^2)}$, then the damping factor is given by:

$$1.8 = 2.0\sqrt{1-\zeta^2}$$

and $\zeta = 0.44$.

(b) Since $\omega t_r = \frac{1}{2}\pi$, then the 100% rise time is given by

$$t_r = \frac{\pi}{2\times 1.8} = 0.87 \text{ s}$$

(c) The percentage overshoot is given by:

$$\%\text{ overshoot} = \exp\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)\times 100\%$$

$$= \exp\left(\frac{-0.44\pi}{\sqrt{1-0.44^2}}\right)\times 100\% = 21\%$$

(d) The 2% settling time is given by:

$$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.44\times 2.0} = 4.5 \text{ s}$$

(e) The number of oscillations occurring within the 2% settling time is given by:

$$\text{number of oscillations} = \frac{2}{\pi}\sqrt{\frac{1}{\zeta^2}-1} = \frac{2}{\pi}\sqrt{\frac{1}{0.44^2}-1} = 1.3$$

Example

The feedback system shown in Figure 4.5 has the transfer function of the forward path as $K/[s(s + a)]$ and the transfer function of the feedback path as 1. What will be the effect on the system response of changing the gain K?

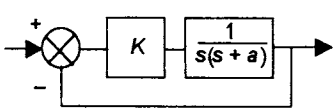

Figure 4.5 *Example*

The transfer function of the closed loop system is:

$$G(s) = \frac{K/[s(s+a)]}{1+K/[s(s+a)]} = \frac{K}{s(s+a)+K} = \frac{K}{s^2+as+K}$$

By comparison with the standard form of the transfer function for a second order system, we have $\omega_n = \sqrt{K}$ and $\zeta = a/2\sqrt{K}$. Thus the rise time is given by $\omega t_r = \frac{1}{2}\pi$, with $\omega = \omega_n\sqrt{(1-\zeta^2)}$, and so is:

$$t_r = \frac{\pi}{2\sqrt{K}\sqrt{1-(a^2/4K)}}$$

The rise time thus decreases as K increases. Thus, increasing the gain decreases the rise time and so increases the speed of response of the system.

The percentage overshoot is given by:

$$\% \text{ overshoot} = \exp\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right) \times 100\%$$

$$= \exp\left(\frac{-(a/2\sqrt{K})\pi}{\sqrt{1-(a^2/4K)}}\right) \times 100\%$$

$$= \exp\left(\frac{a\pi}{\sqrt{4K-a^2}}\right) \times 100\%$$

Thus, increasing K results in an increase in the percentage overshoot.

The 2% settling time is given by:

$$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{a/2} = \frac{8}{a}$$

Thus, the settling time is independent of the gain K.

4.4 Stability

We can define a system as being a *stable system* if, when given an input or a change in input, it has transients which die away with time and leave the system in a steady state condition. The system would be *unstable* if the transients did not die away with time but grew with time and so steady state conditions were never reached.

Consider a second order system with the transfer function:

$$G(s) = \frac{1}{(s+1)(s+2)}$$

The values of s which make the transfer function infinite are termed the *poles* of the system. Thus, the above system has the poles $s = -1$ and $s = -2$. A unit step input to such a system gives an output $Y(s)$:

$$Y(s) = \frac{1}{s(s+1)(s+2)} = \frac{1}{2s} - \frac{1}{s+1} + \frac{1}{2(s+2)}$$

This varies with time as:

$$y = \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t}$$

Each of the poles gives rise to a transient term. Both the resulting exponential terms die away with time to give a steady state value of 0.5,

the more negative the value of s for a pole the more rapidly the corresponding term dies away. Thus, the system is stable.

Now consider a second order system with the transfer function:

$$G(s) = \frac{1}{(s-1)(s-2)}$$

This system has the poles $s = +1$ and $s = +2$. A unit step input to such a system gives an output $Y(s)$:

$$Y(s) = \frac{1}{s(s-1)(s-2)} = \frac{1}{2s} - \frac{1}{s-1} + \frac{1}{2(s-2)}$$

This varies with time as:

$$y = \tfrac{1}{2} - e^{+t} + \tfrac{1}{2}e^{+2t}$$

Each of the positive poles gives rise to exponential terms which grow with time, the larger the value of s for a pole the more rapidly the correspondiong term grows. Thus, the transients do not die away but increase and so the system is unstable.

In general, if a system has a transfer function with a pole which is negative then it gives rise to a transient which dies away with time, whereas if it has a pole which is positive then the transient grows with time. Thus, if a transfer function has a pole which is positive then it is unstable.

In general, for a second order system we have the transfer function (see Section 3.5):

$$G(s) = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{k\omega_n^2}{(s+p_1)(s+p_2)}$$

The roots of the quadratic denominator, i.e. the poles, are given by:

$$p_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$$

$$p_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$$

With $\zeta > 1$ we have real roots then the square root is of a positive quantity and thus the overall root can be written as a real number, as in the examples given above when we had $s = +1$, $s = +2$, $s = -1$ and $s = -2$. With $\zeta < 1$ then the square root is of a negative quantity. If we write j for the square root of minus 1 then the roots can be written as:

$$p_1 = -\zeta\omega_n + j\omega_n\sqrt{1 - \zeta^2}$$

$$p_2 = -\zeta\omega_n - j\omega_n\sqrt{1 - \zeta^2}$$

We can thus write the values of the roots, and so poles, in the form $a + jb$; the jb part of the value is known as an imaginary number. Such

values give rise to oscillatory transients. However, the same rule applies for stability, namely that if the a term is negative then the system is stable and if it is positive it is unstable.

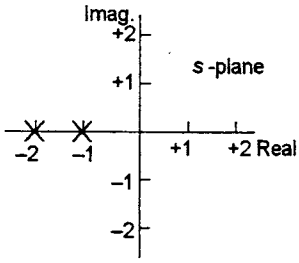

Figure 4.6 *Poles at –1 and –2 for a stable system*

4.4.1 The *s* plane

We can plot the positions of the poles on a graph with the real part of the pole value as the x-axis and the imaginary part as the y-axis. The resulting graph describes what is termed the *s-plane*.

As an illustration, Figure 4.6 shows the s-plane for the transfer function:

$$G(s) = \frac{1}{(s+1)(s+2)}$$

with $s = -1$ and $s = -2$, there being no imaginary terms. This describes an overdamped system. Figure 4.7 shows the s-plane for the transfer function:

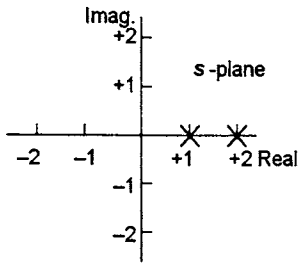

Figure 4.7 *Poles at +1 and +2 for an unstable system*

$$G(s) = \frac{1}{(s-1)(s-2)}$$

with $s = +1$ and $s = +2$, there being no imaginary terms. For the transfer function:

$$G(s) = \frac{1}{s^2 + 2s + 4}$$

we have the roots of the quadratic given by:

$$s = \frac{-2 \pm \sqrt{4-16}}{2} = -1 \pm \mathrm{j}1.73$$

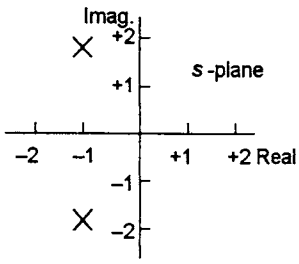

Figure 4.8 *Poles at –1 ± j1.73 for a stable system*

Figure 4.8 shows the s-plane for this transfer function, this describing an underdamped system. For the transfer function:

$$G(s) = \frac{1}{s^2 - 2s + 4}$$

we have the roots of the quadratic given by:

$$s = \frac{2 \pm \sqrt{4-16}}{2} = +1 \pm \mathrm{j}1.73$$

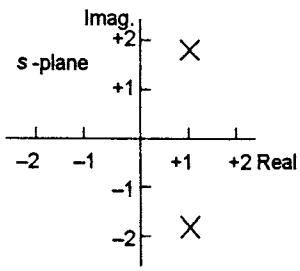

Figure 4.9 *Poles at +1 ± j1.73 for an unstable system*

Figure 4.9 shows the s-plane for this transfer function. For the transfer function:

$$G(s) = \frac{1}{(s+1)^2}$$

we have the roots $s = -1$ and $s = -1$. This is critical damping. Figure 4.10 shows the s-plane for this transfer function.

In general we can state (Figure 4.11):

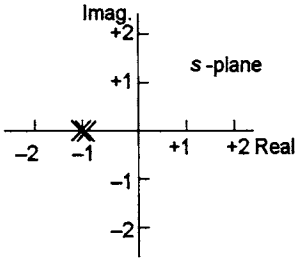

Figure 4.10 *Double pole at −1 for critical damping*

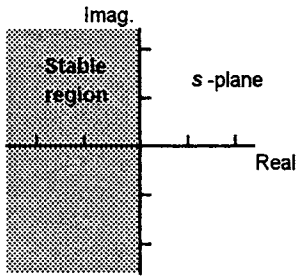

Figure 4.11 *The s-plane: stability when poles are in the left half*

A system is stable if all the system poles lie in the left half of the s-plane.

The relationship between the location of a pole and the form of transient is shown in Figure. 4.12. The more negative the real part of the pole the more rapidly the transient dies away. The larger the imaginary part of the pole the higher the frequency of the oscillation. A system having a pole which has a positive real part is unstable.

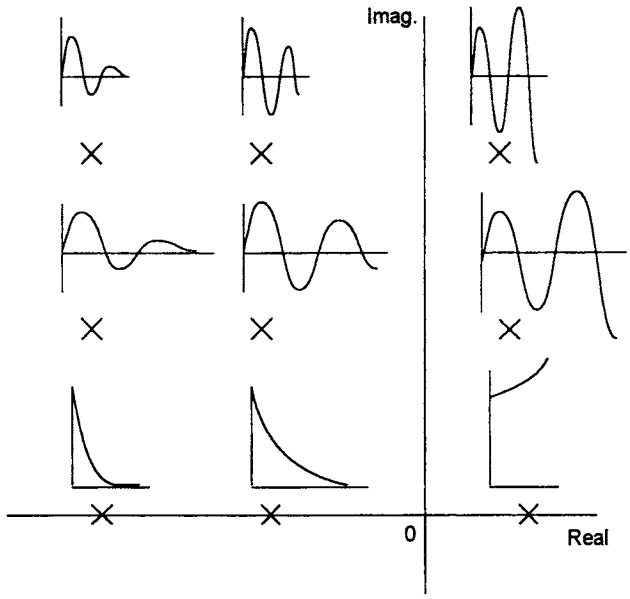

Figure 4.12 *Relationship between pole location and the resulting transient: each oscillatory transient arises from a pair of roots $a \pm jb$ with only one of them shown in the figure*

Example

Which of the following systems are stable: (a) $G(s) = 1/(s^2 + s + 1)$, (b) $G(s) = 1/(s^2 - 5s + 4)$, (c) $G(s) = 1/(s^2 - 2s + 3)$?

(a) This has poles of :

$$s = \frac{-1 \pm \sqrt{1-4}}{2} = -0.5 \pm j1.73$$

The poles will lie in the left half of the s-plane and so the system is stable.

(b) This has poles of $s = +4$ and $s = +1$ and so the poles lie in the right half of the s-plane and the system is unstable.

(c) This has poles of:

$$s = \frac{2 \pm \sqrt{4-12}}{2} = +1 \pm j2.8$$

The poles will lie in the right half of the s-plane and so the system is unstable.

Problems

1 Determine the delay time and the rise time for the following first order systems: (a) $G(s) = 1/(4s + 1)$, (b) $G(s) = 5/(s + 1)$, (c) $G(s) = 2/(s + 3)$.

2 A first order system has a time constant of 30 s. What will be its delay time and rise time when subject to a unit step input?

3 A first order system when subject to a unit step input rises to 90% of its steady state value in 20 s. Determine its time constant, delay time and rise time?

4 Determine the natural angular frequency, the damping factor, the rise time, percentage overshoot and 2% settling time for systems with the following transfer functions: (a) $100/(s^2 + 4s + 100)$, (b) $49/(s^2 + 4s + 49)$.

5 Determine the natural angular frequency, the damping factor, the rise time, percentage overshoot and 2% settling time for a system where the output y is related to the input x by the differential equation:

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 16y = 16x$$

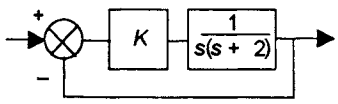

Figure 4.13 *Problem 6*

6 For the feedback system shown in Figure 4.13, what gain K should be used to give a rise time of 2 s?

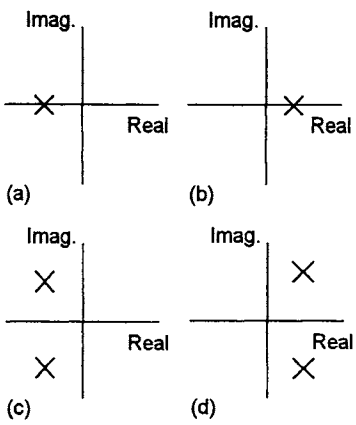

Figure 4.14 *Problem 7*

7 For the poles shown on the s-planes in Figure 4.14, which will give rise to stable transients and which to oscillating transients?

8 Are the systems with the following transfer functions stable?

$$\text{(a)}\ \frac{1}{s^2+2s+1},\ \text{(b)}\ \frac{1}{s^2-2s+10},\ \text{(c)}\ \frac{1}{(s+1)(s-3)}$$

9 Figure 4.15 shows a feedback control system with unity feedback. Will the system be stable when (a) $K = 1$, $G(s) = 1/[s(s + 1)]$, (b) $K = 3$, $G(s) = 1/[s + 4)(s - 1)]$, (c) $K = 5$, $G(s) = 1/[s + 4)(s - 1)]$?

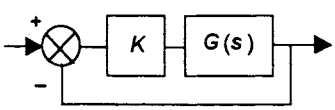

Figure 4.15 *Problem 9*

10 State if the following systems are stable, the relationship between input x and output y being described by the differential equations :

$$\text{(a)}\ \frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x,\ \text{(b)}\ \frac{d^2y}{dt^2} + \frac{dy}{dt} - 6y = x$$